

- What are carbenes? Define the types of carbenes depending on their spin state. Which is more stable between singlet and triplet carbene and why?
- Write the general reaction for the Suzuki-Miyaura cross-coupling reaction and explain each step of the mechanism.
- Write reaction of Fries rearrangement and explain its mechanism.
- Arrange the given compounds in decreasing order of stability / acidic strength. (*Structures provided in exam paper*)
- Write short notes on the following: (i) Electromeric effect (ii) Allenes.
- What is atropisomerism?
- Explain Favorskii rearrangement with its mechanism.
- Write mechanism of Wagner-Meerwein rearrangement.
- Write short notes on the following: (i) Allenes (ii) Spiranes.
- What are spiro compounds (spiranes)? Define with an example.
- Define circular dichroism (CD). Give a schematic diagram of a CD instrument and mention its uses.
- How to differentiate between chiral and achiral molecule? What are the differences?
- Assign the E or Z configuration to the given compounds. (*Structures provided in exam paper*)
- How to differentiate enantiomers and diastereomers? Explain with examples. How do they differ in terms of physical and chemical properties?
- What are enantiomers? Explain with one example and write the properties of enantiomers.
- What is optical activity? How is it measured, and what does it indicate about a compound?
- Explain the principle of optical activity with the help of a schematic diagram of a polarimeter.

Unit 5: Polymers, Water Analysis & Solid Waste Management

- Define functionality and degree of polymerization (DP).
- Explain classification of polymers based on number of monomers and methods of synthesis (sources and structures) with suitable examples.
- Explain zeolite process of softening of water and also write its advantages.

- Calculate temporary and permanent hardness of a water sample containing $\text{Mg}(\text{HCO}_3)_2 = 146 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 324 \text{ mg/L}$, $\text{MgCl}_2 = 190 \text{ mg/L}$ and $\text{CaSO}_4 = 272 \text{ mg/L}$.
- Explain units of hardness of water.
- Calculate temporary, permanent and total hardness of a sample of water containing: $\text{Ca}(\text{HCO}_3)_2 = 45.6 \text{ mg/L}$, $\text{Mg}(\text{HCO}_3)_2 = 25.5 \text{ mg/L}$, $\text{MgCl}_2 = 11.5 \text{ mg/L}$, $\text{CaSO}_4 = 16.5 \text{ mg/L}$, $\text{KCl} = 20.0 \text{ mg/L}$ and $\text{Na}_2\text{SO}_4 = 22.5 \text{ mg/L}$.
- What are the major sources of urban solid wastes?
- Discuss hot lime soda process of softening of water and write advantages and disadvantages of lime soda process.
- Differentiate between thermoplastic and thermosetting polymers with examples.
- Explain Ion exchange process of softening of water and also write its advantages.
- Calculate the temporary and total hardness of a water sample containing $\text{Mg}(\text{HCO}_3)_2 = 85 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 170 \text{ mg/L}$, $\text{MgCl}_2 = 90 \text{ mg/L}$ & $\text{CaSO}_4 = 146 \text{ mg/L}$ (Atomic weight of Mg and Ca are 24 and 40 respectively).
- Write advantages and disadvantages of LS process of softening of hard water.
- Calculate temporary and total hardness of water sample containing: $\text{Mg}(\text{HCO}_3)_2 = 730 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 162 \text{ mg/L}$, $\text{MgCl}_2 = 950 \text{ mg/L}$ and $\text{CaSO}_4 = 136 \text{ mg/L}$.
- Explain three 'R's rules for effective solid waste management.
- What is hardness? Explain types of hardness with example. Why do we express hardness in terms of CaCO_3 equivalents?
- Explain the composition and reason of hardness. Write various units of hardness expression.

Laboratory Concepts & Core Basics

- Define the types of titration.
- Write the example of acidic and basic buffer solution.
- Define the primary and secondary standard reagent with the help of example.
- Why are two indicators used in acid-base titration? / Why do we use two indicators in alkali mixture titration with HCl? Write their pH range.
- Write the Poiseuille's equation.

- Define viscosity. Write the factors affecting viscosity and its unit.
- What is an indicator? Give classification of indicator with suitable examples.
- What is the working pH range of phenolphthalein and methyl orange?
- What is complexometric titration? Write the reaction involved in estimation of hardness of water using EDTA.
- Define normality and molarity of the solution.
- What is buffer? Write an example of acidic & basic buffer.
- What is redox titration? Write the reaction involved in the estimation of iron content using $\text{K}_2\text{Cr}_2\text{O}_7$ as an intermediate and N-phenyl anthranilic acid as an internal indicator.